

D699

# Elementary Mathematics Curriculum

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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## How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

# Items

1.

A third-grade class, which collected 4,382 cans for a food drive run by the PTA, is asked: "What is the value of the 3 in 4,382?" The drive lasted two weeks. What is the correct answer?

- A. 30
- B. 300
- C. 3
- D. Hundreds

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2.

To find  $6 \times 9$  during a lesson taught by a student teacher on her first day, a student writes  $6 \times (4 + 5) = 24 + 30 = 54$ . The student usually sits near the window. Which property did the student use?

- A. Commutative property
- B. Identity property
- C. Associative property
- D. Distributive property

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3.

Two friends who both play on the same soccer team each ate part of identical pizzas. Mia ate  $\frac{3}{8}$  of hers and Jordan ate  $\frac{3}{5}$  of his. The pizzas were ordered from the same shop on a Friday. Who ate more pizza?

- A. It cannot be known without the pizza size.
- B. They ate the same, because both ate 3 pieces.
- C. Jordan, because fifths are larger pieces than eighths.
- D. Mia, because 8 is greater than 5.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

4.

On a morning warm-up projected on the class's new smartboard, second graders solve  $8 + 4 = \underline{\quad} + 5$ . The class has been practicing addition facts to 20 for three weeks. What number belongs in the blank?

- A. 7
- B. 20
- C. 17
- D. 12

|                             |                                           |                       |                                                                                                 |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>____ | <input type="checkbox"/> sure <input type="checkbox"/> unsure <input type="checkbox"/><br>guess |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|

5. [TRANSFER — reversed stem / different variable]

Which student explanation shows a correct understanding of the equal sign?

- A. "It only works when the answer is on the right."
- B. "It means both sides have the same amount."
- C. "It means the answer comes next."
- D. "It means add everything you see."

|                             |                                           |                       |                                                                                                 |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>____ | <input type="checkbox"/> sure <input type="checkbox"/> unsure <input type="checkbox"/><br>guess |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|

6.

A fourth-grade teacher holds up a tile from the cafeteria floor that has four right angles and four equal sides. The tile is blue and was made in a local factory. Which list gives names that all correctly describe it?

- A. Square only
- B. Square, rectangle, rhombus, parallelogram, quadrilateral
- C. Square, rectangle, trapezoid, pentagon
- D. Square and quadrilateral, but not rectangle

|                             |                                           |                       |                                                                                                 |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>____ | <input type="checkbox"/> sure <input type="checkbox"/> unsure <input type="checkbox"/><br>guess |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|

7. [TRANSFER — reversed stem / different variable]

Which statement is always true?

- A. Every parallelogram is a rectangle.
- B. Every rectangle is a square.
- C. Every rhombus is a square.
- D. Every square is a rhombus.

|                             |                                           |                       |                                                                                                 |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>____ | <input type="checkbox"/> sure <input type="checkbox"/> unsure <input type="checkbox"/><br>guess |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|

8.

A class is building a rectangular garden bed 8 feet long and 5 feet wide next to the school's bike rack. The principal donated a bag of soil. How many feet of fencing are needed to go all the way around it?

- A. 80 feet
- B. 13 feet
- C. 40 feet
- D. 26 feet

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

9. [TRANSFER — reversed stem / different variable]

A rectangle has an area of 36 square units and a length of 12 units. What is its perimeter?

- A. 30 units
- B. 15 units
- C. 48 units
- D. 36 units

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

10.

Five students, who all ride the same bus, reported how many books they read this summer: 2, 3, 3, 4, and 18. The student who read 18 goes to the library every day. Which value best describes the typical number of books?

- A. The range, 16 books
- B. 18 books
- C. The median, 3 books
- D. The mean, 6 books

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

11. [TRANSFER — reversed stem / different variable]

The mean of five test scores is 80. Four of the scores are 78, 85, 90, and 72. What is the fifth score?

- A. 405
- B. 81.25
- C. 75
- D. 80

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

12.

A bag from the class game closet holds 3 red, 5 blue, and 2 green cubes. The bag has a torn corner. A student draws one cube without looking. What is the probability it is blue?

- A. 1/2
- B. 1/3
- C. 5/3
- D. 5/5

|                             |                                           |                       |                                                                                                 |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>____ | <input type="checkbox"/> sure <input type="checkbox"/> unsure <input type="checkbox"/><br>guess |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer B Concept 1

A third-grade class, which collected 4,382 cans for a food drive ~~run by the PTA~~, is asked: "What is the value of the 3 in 4,382?" ~~The drive lasted two weeks.~~ What is the correct answer?

**Target word:** value of the digit **Decides:** 3 is in the hundreds place.

|          |                   |                                               |
|----------|-------------------|-----------------------------------------------|
| A        | Confusion set     | Treats the 3 as if it were in the tens place. |
| <b>B</b> | <b>Correct</b>    | $3 \times 100 = 300$ .                        |
| C        | Dropped element   | Names the digit and drops its place.          |
| D        | Adjacent category | Names the place, not the value.               |

## 2. Answer D Concept 2

To find  $6 \times 9$  during a lesson ~~taught by a student teacher on her first day~~, a student writes  $6 \times (4 + 5) = 24 + 30 = 54$ . ~~The student usually sits near the window.~~ Which property did the student use?

**Target word:** name the property **Decides:** A factor multiplied across each part of a sum.

|          |                   |                                                    |
|----------|-------------------|----------------------------------------------------|
| A        | Confusion set     | Commutative only changes order.                    |
| B        | Dropped element   | Identity involves multiplying by 1; no 1 appears.  |
| C        | Adjacent category | Associative changes grouping within one operation. |
| <b>D</b> | <b>Correct</b>    | $6 \times (4 + 5) = 6 \times 4 + 6 \times 5$ .     |

## 3. Answer C Concept 3

Two friends ~~who both play on the same soccer team~~ each ate part of identical pizzas. Mia ate  $\frac{3}{8}$  of hers and Jordan ate  $\frac{3}{5}$  of his. ~~The pizzas were ordered from the same shop on a Friday.~~ Who ate more pizza?

**Target word:** compare fractions **Decides:** Same numerator; fifths are larger pieces than eighths.

|          |                    |                                             |
|----------|--------------------|---------------------------------------------|
| A        | Camouflage         | The stem says the pizzas are identical.     |
| B        | Dropped element    | Drops the size of the pieces.               |
| <b>C</b> | <b>Correct</b>     | $\frac{3}{5} = 0.6 > \frac{3}{8} = 0.375$ . |
| D        | Inverted operation | A larger denominator means smaller pieces.  |

## 4. Answer A Concept 4

On a morning warm-up ~~projected on the class's new smartboard~~, second graders solve  $8 + 4 = \underline{\quad} + 5$ . The class has been practicing addition facts to 20 ~~for three weeks~~. What number belongs in the blank?

**Target word:** fill the blank **Decides:** Both sides must equal 12.

|          |                    |                                                |
|----------|--------------------|------------------------------------------------|
| <b>A</b> | <b>Correct</b>     | $7 + 5 = 12 = 8 + 4$ .                         |
| B        | Camouflage         | Pulls the 20 from the practice detail.         |
| C        | Inverted operation | Adds all three numbers instead of balancing.   |
| D        | Dropped element    | Treats = as "the answer is," dropping the + 5. |

### 5. Answer B Concept 4 • transfer

Which student explanation shows a correct understanding of the equal sign?

**Target word:** pick the relational explanation **Decides:** Equal means the same value on both sides.

|          |                    |                                                          |
|----------|--------------------|----------------------------------------------------------|
| A        | Absolute language  | Equations can be written either way, like $12 = 7 + 5$ . |
| <b>B</b> | <b>Correct</b>     | That is the relational meaning.                          |
| C        | Confusion set      | The operational misconception.                           |
| D        | Inverted operation | Turns the sign into an operation.                        |

### 6. Answer B Concept 5

A fourth-grade teacher holds up a tile from the cafeteria floor that has four right angles and four equal sides. ~~The tile is blue and was made in a local factory.~~ Which list gives names that all correctly describe it?

**Target word:** all correct names **Decides:** A square meets the definitions of rectangle and rhombus.

|          |                   |                                                            |
|----------|-------------------|------------------------------------------------------------|
| A        | Absolute language | Ignores the hierarchy; a square also fits broader classes. |
| <b>B</b> | <b>Correct</b>    | A square has all the properties those classes require.     |
| C        | Confusion set     | A pentagon has five sides.                                 |
| D        | Dropped element   | Four right angles make it a rectangle.                     |

### 7. Answer D Concept 5 • transfer

Which statement is always true?

**Target word:** direction of the hierarchy **Decides:** Every square is a rhombus; not every rhombus is a square.

|          |                    |                                                         |
|----------|--------------------|---------------------------------------------------------|
| A        | Adjacent category  | Parallelograms need not have right angles.              |
| B        | Confusion set      | Rectangles need not have equal sides.                   |
| C        | Inverted operation | A rhombus need not have right angles.                   |
| <b>D</b> | <b>Correct</b>     | A square has four equal sides, the rhombus requirement. |

### 8. Answer D Concept 6

A class is building a rectangular garden bed 8 feet long and 5 feet wide ~~next to the school's bike rack. The principal donated a bag of soil.~~ How many feet of fencing are needed to go all the way around it?

**Target word:** distance around **Decides:** Fencing = perimeter:  $2 \times (8 + 5)$ .

|          |                    |                                               |
|----------|--------------------|-----------------------------------------------|
| A        | Inverted operation | Doubles the area instead of adding the sides. |
| B        | Dropped element    | Adds only two sides.                          |
| C        | Confusion set      | That is the area in square feet.              |
| <b>D</b> | <b>Correct</b>     | $8 + 5 + 8 + 5 = 26$ .                        |

### 9. Answer A Concept 6 • transfer

A rectangle has an area of 36 square units and a length of 12 units. What is its perimeter?

**Target word:** solve for width, then perimeter **Decides:** Width =  $36 \div 12 = 3$ .

|          |                   |                                                  |
|----------|-------------------|--------------------------------------------------|
| <b>A</b> | <b>Correct</b>    | Width 3; $2 \times (12 + 3) = 30$ .              |
| B        | Dropped element   | Adds length and width once, not twice.           |
| C        | Adjacent category | Adds $36 + 12$ , mixing area with a side length. |
| D        | Wrong stem number | That is the area.                                |

### 10. Answer C Concept 7

Five students, ~~who all ride the same bus~~, reported how many books they read this summer: 2, 3, 3, 4, and 18. The student who read 18 ~~goes to the library every day~~. Which value best describes the typical number of books?

**Target word:** typical value with an outlier **Decides:** 18 is an outlier; the median resists it.

|   |                   |                                                           |
|---|-------------------|-----------------------------------------------------------|
| A | Confusion set     | Range measures spread, not typical value.                 |
| B | Camouflage        | The frequent library visitor is the outlier, not typical. |
| C | <b>Correct</b>    | Middle of 2, 3, 3, 4, 18 is 3.                            |
| D | Adjacent category | $30 \div 5 = 6$ is pulled up by the outlier.              |

### 11. Answer C Concept 7 • transfer

The mean of five test scores is 80. Four of the scores are 78, 85, 90, and 72. What is the fifth score?

**Target word:** solve for the missing value **Decides:** Total must be  $5 \times 80 = 400$ .

|   |                    |                                                                          |
|---|--------------------|--------------------------------------------------------------------------|
| A | Inverted operation | Adds $325 + 80$ instead of subtracting 325 from the needed total of 400. |
| B | Dropped element    | $325 \div 4$ is the mean of the four, not the fifth score.               |
| C | <b>Correct</b>     | $400 - (78 + 85 + 90 + 72) = 400 - 325 = 75$ .                           |
| D | Wrong stem number  | Assumes the missing score equals the mean.                               |

### 12. Answer A Concept 8

A bag ~~from the class game closet~~ holds 3 red, 5 blue, and 2 green cubes. ~~The bag has a torn corner.~~ A student draws one cube without looking. What is the probability it is blue?

**Target word:** P(blue) **Decides:** 5 blue out of 10 total.

|   |                    |                                                                |
|---|--------------------|----------------------------------------------------------------|
| A | <b>Correct</b>     | $5 \div 10 = 1/2$ .                                            |
| B | Adjacent category  | Divides by the number of colors, which are not equally likely. |
| C | Wrong stem number  | Divides by the red count; probabilities cannot exceed 1.       |
| D | Inverted operation | Divides by the non-blue count instead of the total.            |

## Triage grid

Write each item number in the box it belongs to.

|                       | Right                                                 | Wrong                                                                                 |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owned.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| Trap               | What it does                                                 | In this drill | Your misses |
|--------------------|--------------------------------------------------------------|---------------|-------------|
| Dropped element    | Leaves out a required term or step.                          | 8             |             |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 3             |             |
| Adjacent category  | The right idea in the wrong bucket.                          | 6             |             |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 7             |             |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 2             |             |
| Confusion set      | Every option is real; the test is exact role-matching.       | 7             |             |
| Camouflage         | Answers a stem fact that was never the question.             | 3             |             |

# Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                                | Rule                                                                                                                                                        | Trap                                                              | Tell                                                            |
|---|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------|
| 1 | <b>Place value: digit vs. value</b>    | A digit's value is the digit times its place (in 4,382 the 3 is worth 300); each place is ten times the place to its right.                                 | Naming the digit or the place instead of the value.               | "What is the value of the digit...?"                            |
| 2 | <b>Properties of operations</b>        | Commutative changes order, associative changes grouping, distributive splits a factor across a sum: $a(b + c) = ab + ac$ .                                  | Calling any regrouping "commutative."                             | Parentheses moving, or a factor multiplied across a sum.        |
| 3 | <b>Comparing fractions</b>             | With the same numerator, the smaller denominator is the larger fraction (bigger pieces); compare unlike fractions with a common denominator or a benchmark. | Thinking a bigger denominator means a bigger fraction.            | Two fractions with the same top number.                         |
| 4 | <b>The equal sign is relational</b>    | The equal sign means both sides have the same value, not "write the answer next."                                                                           | Filling the blank in $8 + 4 = \underline{\quad} + 5$ with 12.     | An operation on both sides of the equal sign.                   |
| 5 | <b>Classifying shapes by hierarchy</b> | Shapes are classified by properties, so a square is also a rectangle, a rhombus, a parallelogram, and a quadrilateral; the reverse is not always true.      | Saying a square is not a rectangle.                               | "Which names describe...?" or "always/sometimes/never."         |
| 6 | <b>Area vs. perimeter</b>              | Perimeter is the distance around (add the sides); area is the space covered (length $\times$ width, in square units).                                       | Using area for fencing or trim, or perimeter for carpet or tile.  | Fence, border, trim (perimeter) vs. cover, carpet, tile (area). |
| 7 | <b>Mean vs. median</b>                 | The mean is the total divided by the count and is pulled by outliers; the median is the middle value and resists them.                                      | Reporting the mean as "typical" when one value is extreme.        | A data set with one very large or small value.                  |
| 8 | <b>Simple probability</b>              | Probability = favorable outcomes $\div$ total equally likely outcomes, a number from 0 to 1.                                                                | Dividing by the number of categories or by the unfavorable count. | A bag, spinner, or number cube with counts.                     |

**Course scope (public description):** Elementary Mathematics Curriculum is a detailed and practical course designed for educators who seek to expand their knowledge of the mathematics curriculum in elementary classrooms. This course provides the essential mathematical concepts suitable for young learners, including numbers, operations, and algebraic thinking, spanning through the domains of geometry, measurement, data, statistics, and probability. Emphasis is placed on engaging with a curriculum that not only aligns with educational standards but also nurtures a love for mathematics. By the end of this course, candidates are provided with the tools and confidence to foster a strong mathematical foundation in their elementary students.